

SMART HOME AUTOMATION SYSTEM WITH HAND GESTURE RECOGNITION

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SEPTEMBER 2025

CERTIFICATION OF THE RESEARCHERS

We certify that the capstone project entitled “**SMART HOME AUTOMATION SYSTEM WITH HAND GESTURE RECOGNITION**”, submitted to the College of Computer Studies, Eastern Samar State University (Guiuan Campus), is our own independent work that has not been presented and submitted to another university or faculty.

We, furthermore, cede of copyright of this capstone project in favor of Eastern Samar State University.

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SMART HOME AUTOMATION SYSTEM WITH HAND GESTURE RECOGNITION

**An Undergraduate Capstone Project Presented to the
Faculty of the College of Computer Studies
Eastern Samar State University
Guiuan, Eastern Samar**

**In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Technology**

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September 2025

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ABSTRACT

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This research presents an innovative gesture-controlled smart home automation system designed to enhance accessibility and usability for individuals with mobility impairments, speech difficulties, and other accessibility needs. Traditional smart home interfaces often rely on voice commands or mobile applications, which can be restrictive for the elderly, persons with disabilities, or users unfamiliar with such technologies. Although gesture-based systems provide a more inclusive alternative, many existing solutions lack adaptability and require users to memorize predefined gestures.

The proposed system employs computer vision and machine learning to enable real-time recognition of hand gestures for contactless control of lighting, appliances, and security systems. It supports customizable gesture configurations, allowing users to define personalized controls based on their comfort and preferences. By integrating advanced technologies such as FastAPI and WebSocket, the system ensures high-speed, real-time responsiveness, delivering near-instant feedback and seamless interaction.

Moreover, the system supports offline functionality and performs reliably under various indoor conditions. This research contributes to the advancement of smart home technology by introducing a customizable, intuitive, and high-performance control method that enhances accessibility, adaptability, and overall user experience.

Keywords: Smart Home Automation System, Gesture-based, Computer Vision, Machine Learning.

Chapter I

INTRODUCTION

The rapid advancement of smart home technologies has revolutionized modern living by enhancing convenience, security, and energy efficiency. Current systems predominantly rely on mobile applications and voice commands, enabling users to manage appliances, lighting, and security remotely (Sharkawy, Hasanin, Sharf, Mohamed, & Elsheikh, 2022). However, these conventional interfaces present significant accessibility challenges for individuals with mobility or speech impairments, as well as in situations where voice or touch interactions are impractical. This underscores the necessity for alternative control mechanisms that prioritize inclusivity and ease of use (Yu, Ouyang, Wang, Tao, & Jing, 2022).

Existing research in home automation has largely centered on voice recognition and smartphone-based applications, which have proven effective in improving user convenience (Ok, 2025). Yet, these methods fall short in addressing accessibility barriers, as voice systems may struggle with speech variations or background noise, while touch interfaces assume full motor control. Gesture-based interaction has emerged as a promising solution, particularly in gaming and virtual reality, offering an intuitive and contactless means of engaging with technology (Bankar, Harale, & Karande, 2021).

Recent innovations in smart home technology, such as Bluetooth-enabled Arduino systems and sensor-driven interfaces, have demonstrated success in automating tasks and reducing energy consumption (Sharkawy et al., 2022). However, these solutions often depend on smartphones, which may not be suitable for users with physical impairments. Studies by Rustam et al. (2023) highlight the potential of gesture recognition, achieving 94% accuracy with inertial sensors, thereby validating its feasibility for non-contact home automation. Moreover, gesture recognition has

been widely explored for health monitoring applications, further emphasizing its potential for inclusive smart home environments (Rustam et al., 2023).

Smart home automation systems that incorporate gesture-based control methods can enhance accessibility by minimizing reliance on voice commands and touch interactions. However, many existing solutions require additional wearable devices or extensive calibration, limiting their practicality in real-world home settings. To address this limitation, gesture recognition systems leveraging computer vision can provide a seamless and intuitive way to control home automation features without the need for physical contact or auxiliary hardware. Despite the advancements in smart home technology, a gap remains in the development of intuitive, gesture-based control systems specifically tailored for home automation. While some gesture-controlled systems exist, they often rely on external hardware or require complex calibration, making them less practical for everyday users. Furthermore, deep learning-based gesture recognition has demonstrated significant potential in improving smart home interactions by eliminating the need for traditional interfaces (Bankar et al., 2021).

This study aims to design a gesture-controlled home automation system leveraging computer vision and machine learning. The system will feature real-time gesture recognition, customizable commands, and AI-driven adaptability to user-specific gestures. By bypassing the need for voice or touch inputs, the proposed system seeks to enhance accessibility for individuals with disabilities or situational constraints. Prior research underscores the potential of integrating artificial intelligence (AI) and the Internet of Things (IoT) to create more inclusive smart home environments (Ok, 2025).

The significance of this research lies in its contribution to inclusive technology. By offering a contactless interaction method, the system empowers users with disabilities, fostering independence and improving quality of life. Its adaptability to diverse lighting conditions and user preferences ensures reliable performance across environments. This study aligns with the growing emphasis on accessible design,

particularly for aging populations and individuals with physical limitations (Yu et al., 2022), thereby paving the way for a more inclusive future in smart home automation.

Objectives of the Study

The general objective of this study was to design, develop, and evaluate a smart home automation system with hand gesture recognition that enhances accessibility, particularly for individuals with mobility and speech impairments. Specifically, it aimed to:

1. design and develop a smart home automation system with hand gesture recognition to enhance smart home accessibility, particularly for users with mobility and speech impairments;
2. evaluate the system through benchmark, pilot, and final testing in terms of the following criteria:
 - 2.1) Functionality;
 - 2.2) Reliability;
 - 2.3) Usability; and
 - 2.4) Maintainability.
3. assess the usability and effectiveness of the developed system based on user feedback and system performance

Significance of the study

Smart home technology has made remarkable strides in recent years; however, many existing systems continue to exclude users with mobility or speech impairments due to their heavy reliance on voice and touch interfaces. For the **institution**, this research reinforces its commitment to advancing assistive technologies by exploring novel, user-centered interfaces that bridge the accessibility gap.

Within the **college and department**, this study serves as a tangible application of emerging fields such as computer vision, machine learning, and the Internet of

Things (IoT). It not only enriches academic discourse but also provides a practical framework for future interdisciplinary research and innovation.

For the **proponents**, the project presents an opportunity to deepen their technical expertise while developing meaningful solutions that improve daily living for individuals with disabilities.

For the **end users**—particularly the elderly and persons with physical challenges—this work offers a more natural and contactless way to interact with their environment, fostering independence and enhancing quality of life.

Finally, for the **technology industry and smart home developers**, the study's findings advocate a paradigm shift toward inclusive design, encouraging the creation of smarter, more accessible, and human-centered home automation systems.

Scope and Delimitation of the Study

This study focused on the development of a real-time hand gesture recognition system for smart home automation, which utilized computer vision and machine learning to enable contactless control of lighting, appliances, and security systems. The system included a web-based application that allowed gesture customization and monitoring. It was trained using publicly available datasets supplemented with real-time user input to improve recognition accuracy and adaptability. Testing was conducted in controlled indoor environments with participants of varying mobility and speech impairments to evaluate the system's functionality, reliability, and usability.

However, the study was limited to indoor implementation, as outdoor performance was affected by environmental factors such as lighting conditions and background complexity. Voice commands were excluded as a primary mode of interaction, and the initial prototype supported only a predefined set of gestures. Furthermore, system responsiveness varied depending on hardware specifications, particularly when deployed on lower-end devices. These limitations were identified as areas for improvement in future iterations to enhance the system's adaptability, scalability, and overall performance.

Definition of Terms

Computer Vision. A field of artificial intelligence that enabled the system to capture, process, and interpret hand images from a camera. In this study, computer vision was used to detect specific gestures that served as input commands for controlling home appliances.

Functionality. Refers to the system's ability to perform its intended tasks, such as detecting fire and sending SMS alerts to users for safety. In this study, functionality measures how effectively the system executes its operations.

Gesture Recognition. A technique that identified hand movements or positions captured through a camera and converted them into predefined commands. In this system, gesture recognition allowed users to operate smart home devices through hand gestures instead of physical buttons or voice commands.

Home Automation. The integration of technology to control household appliances, lighting, security, and other systems either remotely or autonomously. In this study, home automation served as the core concept behind enabling gesture-based control of home devices.

Human-Computer Interaction (HCI). The study and design of interfaces that allow users to interact effectively and intuitively with technology. In this study, HCI principles guided the creation of an interface that ensured users could control home appliances through gestures in a seamless and user-friendly manner.

Internet of Things (IoT). A network of interconnected devices capable of communicating and exchanging data to automate tasks and improve system functionality. In this project, IoT enabled communication between the gesture recognition system and smart home appliances.

Machine Learning (ML). A subset of artificial intelligence that allowed systems to learn patterns from data and improve their performance without being explicitly programmed. In this study, ML techniques supported the accurate interpretation of hand gestures captured by the camera.

Maintainability. Refers to how easily a system can be modified to correct faults, improve performance, or adapt to changing conditions. As used in this study, maintainability pertains to how easily the smart home system can be updated, repaired, or enhanced to ensure continued reliability and performance.

Reliability. Describes the system's consistency in performing its functions accurately over time, such as reliably detecting hazards and delivering notifications without failure. In this study, reliability measures how consistently the system functions under various testing conditions.

Smart Home. A residence equipped with automated systems that allow remote or autonomous control of appliances, lighting, security, and other household functions. In this research, the smart home concept was implemented through gesture-based interaction to enhance convenience, security, and efficiency.

User Accessibility. The extent to which individuals, including those with physical or cognitive limitations, can effectively use and benefit from a technology. In this study, user accessibility refers to the system's ability to accommodate different users by providing an intuitive, gesture-based interface that does not rely on traditional input devices.

Usability. Refers to how easy and efficient it is for users to interact with the system to accomplish their tasks. It involves the system's clarity, intuitiveness, and user-friendliness, such as providing a simple interface, clear instructions, and smooth navigation. In this study, usability measures how easily users can operate the smart home system, including how intuitive the interface is, how simple it is to access its features, and how efficiently users can complete tasks without confusion or errors.

Chapter II

REVIEW OF RELATED LITERATURE

This chapter provides a review of related literature and studies from various authors that support and offer insights relevant to the present research.

Related Literature

The evolution of smart home automation was significantly influenced by advancements in computer vision, machine learning, and the Internet of Things (IoT). Traditional home automation systems primarily relied on manual controls, mobile applications, and voice commands for operation (Kamil Abdalhameed, 2019). However, these methods posed limitations in accessibility and intuitiveness, particularly for users with disabilities and the elderly (Kamil Abdalhameed, 2019). The integration of gesture recognition in home automation systems presented a promising alternative by enabling hands-free interaction.

Gesture-based smart home automation required simple and intuitive hand gestures to ensure accuracy and accessibility. For instance, an open palm could switch on devices, a closed fist could turn them off, and swiping up or down could adjust brightness, while swiping sideways could change media. Circular finger motions could control light colors, and gestures like a peace sign or crossed arms could unlock or lock doors. For climate control, pushing the palm forward increased speed or temperature, while pulling back decreased it (Hemalatha et al., 2021; Ivanova & Setz, 2021; Tasfia et al., 2024; Ok, 2025).

Gesture recognition had been a topic of extensive research in human-computer interaction (HCI), particularly in enhancing accessibility and user experience. Traditional gesture-based systems required predefined gestures, which were cumbersome and non-intuitive for users (Tasfia, Yusoh, Habib, & Mohaimen, 2024). Computer vision-based gesture recognition, particularly those leveraging depth-sensing cameras and neural networks, demonstrated high accuracy in detecting and

interpreting hand movements (Hemalatha, Ramya, & Professor, 2021). Microsoft's Kinect sensor, for instance, was widely used for full-body tracking and gesture control, offering a structured light application that measured depth and movement (Hemalatha et al., 2021).

Machine learning and artificial intelligence (AI) played crucial roles in enhancing the adaptability and accuracy of gesture recognition systems. AI-driven smart home systems were able to learn user preferences and adapt interactions based on behavior patterns (Ok, 2025). For instance, deep learning-based gesture recognition models trained on large datasets demonstrated superior accuracy in identifying complex gestures compared to rule-based systems (Tasfia et al., 2024). Moreover, studies showed that real-time feedback mechanisms integrated with machine learning models significantly enhanced gesture recognition accuracy over time (Zaro, Tamimi, & Barakat, 2021). Research also emphasized the need for dynamically adaptable automation rules that evolved based on user feedback, reducing the need for manual reconfiguration (Ivanova & Setz, 2021).

The incorporation of IoT in smart home automation further expanded the scope of gesture-based control. IoT-enabled home systems allowed seamless device integration and remote control, ensuring flexibility and convenience (Rajeswari & Rajesh, 2019). Furthermore, smart home hubs such as Google Home and Amazon Alexa increasingly adopted AI and IoT for improved interoperability between devices (Ok, 2025). This integration allowed gesture-controlled automation systems to communicate with other smart devices, enhancing overall efficiency and usability. Additionally, IoT-based home automation solutions were developed using NodeMCU and Wi-Fi modules to facilitate real-time control of home appliances (Raj, Rajput, Singh, & Professor, 2023). These advancements catered to individuals with disabilities and the elderly by offering voice and gesture control as alternatives to traditional switches.

Despite these advancements, challenges remained in the widespread adoption of gesture-controlled home automation. One of the primary concerns was ensuring real-time responsiveness and accuracy in varying environmental conditions (Hemalatha et al., 2021). Low-light environments and occlusions impacted the performance of vision-based gesture recognition systems, necessitating the use of advanced filtering techniques and multimodal sensing (Tasfia et al., 2024). Additionally, privacy and security considerations had to be addressed, as AI-driven smart home systems required continuous monitoring and data processing (Singh et al., 2023). Furthermore, while voice commands served as an alternative means of automation, challenges such as ambient noise, speaker variability, and language support had to be mitigated to ensure robust performance (Azhiimah, Khotimah, Sumbawati, & Santosa, 2020).

In conclusion, adding easy-to-understand hand gestures to smart home automation systems gave users a more accessible, hands-free, and natural way to interact with gadgets. These systems were made more user-friendly and inclusive, particularly for the elderly and people with disabilities, by combining gestures such as swiping for adjustments, an open palm for turning on, a fist for turning off, and distinct signs for security or emergencies. In contemporary smart home environments, gesture-based automation guaranteed efficiency and adaptability while also improving usability.

Related Studies

Home automation rapidly evolved with advancements in the Internet of Things (IoT), artificial intelligence (AI), and smart interfaces. Traditional home automation systems often relied on voice and mobile app-based controls, which were limiting for users with disabilities or those unfamiliar with smart home interfaces. To enhance accessibility, researchers explored various control methods, including voice, gesture, and centralized mobile-based applications.

Gesture-based systems gained traction as they provided an intuitive, hands-free approach to controlling home appliances. Mohamed and El Shenawy (2023) proposed an IoT-based automation system that integrated smart devices and cloud computing, ensuring real-time monitoring and control of home appliances. Another approach by Amin (2022) developed a voice-controlled home automation system that used IoT technology to enable users to manage appliances remotely through mobile applications, leveraging AI for an enhanced user experience.

Voice control remained one of the most commonly implemented solutions in smart homes. Meraj (2021) explored the use of Google Assistant and IFTTT (If This Then That) to create a cost-effective automation system, allowing users to issue voice commands to control various household devices. Similarly, Amin (2022) examined Amazon Alexa's role in smart home automation, providing insights into integrating AI-powered voice assistants for seamless home control. However, voice-controlled systems had inherent limitations, particularly for users with speech impairments or those in noisy environments.

Research on AI-driven voice assistants such as Amazon Alexa demonstrated their potential to enhance smart home automation by offering natural language interactions and real-time adaptability to user preferences (Gao et al., 2023). The Alexa Arena platform, designed to facilitate human-robot interaction, enabled users to engage with AI-powered agents for task automation and smart device control. Studies suggested that AI-powered voice interfaces improved user experience by incorporating contextual awareness, allowing systems to adjust responses based on environmental and user-specific factors (Gao et al., 2023). However, while AI-driven voice assistants significantly enhanced automation, they still posed challenges in understanding complex commands, handling multiple concurrent requests, and ensuring privacy and security in voice data processing.

To further enhance accessibility, systems tailored specifically for elderly and disabled individuals were explored. Lawal, Mohammed, Ibrahim, and Gital (2024)

designed an automation system for elderly and physically challenged users, ensuring ease of use and remote operation while integrating enhanced security features. Murugadhas, Mohammed Al-Aamri, and Sulaiman Al-Sabahi (2021) proposed a mobile-based home automation system that allowed elderly and handicapped individuals to control household appliances without physical interaction.

Recent developments expanded home automation beyond traditional interfaces. Sudhindra Madi (2021) introduced an advanced home automation system using Raspberry Pi, which enhanced real-time control of devices through mobile and web applications, improving accessibility for users who preferred visual interfaces. Similarly, Joseba, Rodríguez, García, and Montoya (2012) developed a centralized home automation system that integrated mobile and web-based controls, allowing users to automate lighting, sensors, and security systems remotely. These advancements highlighted the growing importance of multi-platform integration in smart home systems.

Bluetooth-based automation was also explored as an alternative to IoT and cloud-based solutions. Iyen, Ayomanor, and Orseer (2024) implemented a Bluetooth-enabled home automation system that allowed users to control appliances within a short-range radius, reducing dependency on internet connectivity. This approach was particularly useful for areas with unreliable internet access or for users who prioritized offline control.

Despite advancements in smart home automation, challenges persisted in ensuring system accuracy across varying lighting conditions, diverse user movements, and different environmental settings. Variations in room lighting, background interference, and unpredictable human gestures often affected the precision and responsiveness of automated systems, making consistent performance a key concern. Additionally, offline functionality and enhanced security measures were crucial for preventing unauthorized access, safeguarding user data, and ensuring overall system reliability (Mohamed & El Shenawy, 2023). Addressing these issues was particularly

important for maintaining trust in smart home technologies, especially in households with vulnerable users or in areas with intermittent internet connectivity.

AI-driven voice assistants continued to shape the future of home automation. Meraj (2021) explored the integration of Google Assistant with smart home devices, emphasizing its potential to provide voice-controlled automation tailored to user preferences. This research highlighted the increasing role of AI in personalizing smart home experiences, and also suggested that combining multiple control methods, such as gesture and voice, could create more robust and inclusive systems suitable for diverse users.

Overall, the reviewed studies indicate a clear trend toward multi-modal, machine learning-enhanced home automation systems that prioritize accessibility, convenience, and adaptability. Researchers consistently emphasize the need to integrate multiple interfaces, gesture, voice, mobile, and web, to accommodate different user needs, especially for the elderly and individuals with disabilities. At the same time, challenges such as system accuracy, reliability under varying environmental conditions, offline functionality, and data security remain critical areas for improvement. Together, these findings suggest that the future of home automation lies in systems that are flexible, secure, and capable of personalizing interactions, combining advanced machine learning capabilities with inclusive design principles to meet the diverse requirements of modern households.

Research Framework

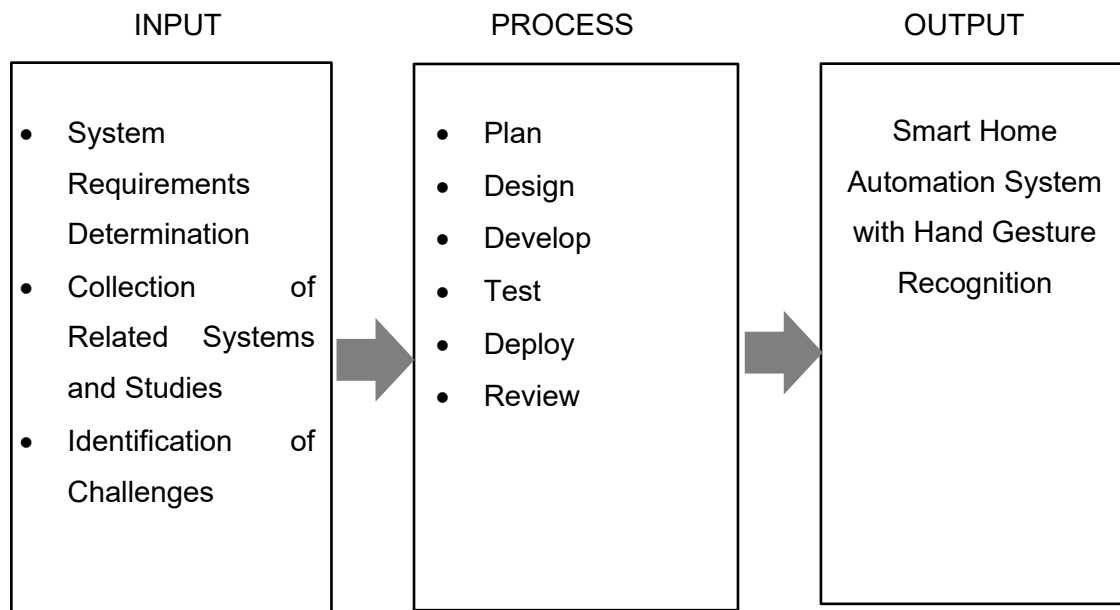


Figure 1. Research Flow

The researchers adopted the Input-Process-Output (IPO) model from the study of Tabuena (2021) to clearly illustrate the overall research flow and ensure an organized presentation of how the system was conceptualized and developed. As shown in Figure 1, the input phase involved identifying the system requirements, reviewing existing systems and related literature, and determining the key challenges and limitations that needed to be addressed. These inputs served as the foundation of the study, guiding the direction of the system's features and functionalities. During the process phase, a series of activities were undertaken, including requirement gathering, system design, prototyping, iterative development, component integration, testing procedures, deployment, and the incorporation of user feedback to continuously refine and improve the system. Finally, the output of the study resulted in the development of the Gesture-Based Smart Home Automation System, together with its Web-Based Smart Home Dashboard, which collectively represent the completed product demonstrating the feasibility and effectiveness of gesture-controlled smart home technology.

Chapter III

METHODOLOGY

This chapter outlined the methodology for developing the Smart Home Automation System with Hand Gesture Recognition. It detailed the research design, software model, participant selection, research instruments, data collection procedures, and analysis methods.

Research Design

The study adopted a developmental research approach to create a location-aware gesture recognition system. This method enabled iterative development with rapid feedback and real-time testing, ensuring continuous refinement of the system.

Software Model



Figure 2. *The Agile Development Model (Source: Google)*

The development of the study used the Agile model. The Agile model was a flexible and iterative approach to software development that emphasized continuous collaboration, adaptability, and incremental progress. Instead of building the entire system at once, Agile focused on delivering small, functional parts of the system in

short cycles called sprints. This method allowed continuous feedback and improvements, ensuring that the final system met user requirements effectively. The Agile model consisted of iterative phases, as shown in Figure 2, where each cycle involved planning, development, testing, and review before moving to the next phase.

Planning. In Agile development, requirements were gathered continuously rather than all at once. The team collaborated with the respondents to identify the most critical features and functionalities. These requirements were documented as user stories in the product backlog and were prioritized for development.

Design. Instead of a fixed design phase, Agile teams planned and designed in small increments. Before each sprint, the team selected high-priority user stories and created a basic design using wireframes, process flows, and system diagrams to guide development.

Development. The selected features were developed during short development cycles called sprints. Developers wrote code, integrated new features, and ensured that each increment was functional and usable. Agile encouraged continuous integration, meaning code was frequently tested and merged.

Testing. Unlike traditional models where testing happens at the end, Agile testing was continuous. Each sprint included unit testing, integration testing, and user acceptance testing to ensure that the functionality worked as expected. Automated testing tools were often used to speed up the process.

Deployment. Agile followed the continuous deployment principle, meaning that working software was released frequently. Instead of waiting for a final version, new updates were deployed to users in small, manageable increments, ensuring faster delivery of value.

Review. At the end of each sprint, a sprint review was conducted where the team presented the developed features to the respondents for feedback. A sprint retrospective followed, where the team discussed what went well, what could be improved, and how to enhance the next sprint.

Maintenance and Continuous Improvement. Agile emphasized continuous feedback and adaptation. Updates, bug fixes, and enhancements were made regularly based on user feedback. This cycle repeated for each sprint, ensuring that the software evolved based on real-world needs.

Locale of the study



Figure 3. *Brgy. Salug (Source: Google Maps)*

The research was conducted in Barangay Salug, a coastal community in Guiuan, Eastern Samar, characterized by its quiet residential areas and typical household environments. The barangay reflects a blend of rural living and gradually emerging access to modern technologies, making it a suitable and realistic setting for testing systems aimed at improving daily household activities. Conducting the study in this locale allowed the researchers to observe how the system performs under real-life conditions within an ordinary community, ensuring that the results accurately represent

practical use rather than controlled or idealized scenarios, while also highlighting how such innovations can support local needs and daily routines.

Respondents of the Study

Table 1. *Distribution of Respondents*

Respondent	Tests	Number of Respondents
Researchers	Benchmark	3
IT Experts	Pilot	30
Homeowners	Final	30
TOTAL		63

The respondents of the study were composed of three groups: the Benchmark Group, which consisted of the researchers; the Pilot Group, which included selected IT experts; and the Final Testing Group, which involved the homeowners.

Research Instrument

A survey questionnaire was used as the research tool in the study. The ISO 25010 adapted questionnaire was used to gather data from the respondents. The survey's outcomes were used to assess the study that the researchers have presented.

The interpretation of the gathered data was conducted using the following values:

Table 2. *Range and Instrument*

Scale	Range	Description	Interpretation
5	4.21 – 5.00	Excellent	Highly Acceptable
4	3.41 – 4.20	Very Good	Acceptable
3	2.61 – 3.40	Good	Moderately Acceptable
2	1.81 – 2.60	Fair	Less Acceptable
1	1.00 – 1.80	Poor	Not Acceptable

Data Gathering Procedure

The researchers first sought formal approval by submitting a written request addressed to the Campus Administrator of Eastern Samar State University, Guiuan Campus, ensuring that the data-gathering procedures followed the institution's protocols and ethical guidelines. Once permission was granted, the researchers proceeded with the distribution of the survey questionnaire, personally administering each copy to the identified respondents to ensure clarity of instructions and immediate engagement. This personal administration also allowed the researchers to address any questions or concerns from participants, helping to ensure accurate and reliable responses. After the questionnaires were completed, the researchers collected all forms on the same day to maintain the timeliness of the data. The gathered information was then carefully organized, tabulated, and subjected to systematic analysis to generate meaningful insights for the study.

Analysis of Data

The collected survey data were analyzed using appropriate statistical methods. The responses were compiled, structured, and systematically interpreted through mean, frequency, and percentage computations to derive meaningful insights.

Developmental Cost

Table 3. Developmental Cost

Category	Cost Components	Description	Estimated Cost (PHP)	Priority (H/M/L)
Software	Development Tools & Licenses	VS Code IDE, Arduino IDE, Figma (all free versions used)	₱0	High (H)
Hardware	Development Devices	ESP32 Dev kit, ESP32-CAM, Relay Module, Contactors	₱3,000	High (H)
	Tools & Equipment	Multimeter, Screwdriver, Drill Bit, Bolts and Nuts, Construction adhesive, L-Brackets, Jumper wires	₱2,500	Medium (M)
	Prototype Setup / Physical Enclosure	Acrylic glass, PVC board, Wiring	₱1,300	High (H)
	Demonstration Devices	Clip Fan, LED Bulb, Extension Cords	₱1,500	High (H)
	Peripheral Devices	Laptop, Smartphone (Personal, testing web app)	₱0	High (H)
Testing	Automated Testing Tools	Postman, ThunderClient	₱0	Medium (M)
Miscellaneous	Documentation & Printing	Project printing, binding	₱2,500	High (H)
	Contingency Buffer	Unexpected costs (10% of total)	₱800-₱1,000	High (H)

This table outlines the hardware and software used in creating the Smart Home Automation System with Hand Gesture Recognition. The development of the smart home system utilized existing tools and resources to minimize expenses while ensuring functionality and efficiency. Most tools and devices are free since they are open-source or personally owned.

Chapter IV

RESULTS AND DISCUSSION

This section provided an analysis, interpretation, and assessment of the system to illustrate its functionality.

The following figures showed the hardware, layouts, and design of the proposed system, “Smart Home Automation System with Hand Gesture Recognition.”

Design and Functionality of the System



Figure 4. *ESP32-CAM*

Figure 4 shows the ESP32-CAM, a small and low-cost development board manufactured by Espressif that combines an ESP32 microcontroller with a camera module. It is widely used in DIY electronics, IoT, and hobby projects because it can handle both onboard processing and Wi-Fi/Bluetooth connectivity, making it suitable for embedded vision applications. In the smart home system, the ESP32-CAM functions as the primary gesture detection unit, capturing live video input and transmitting it to the server for processing, enabling real-time recognition of hand gestures that trigger automated device control.



Figure 5. *ESP32 Development Board*

Figure 5 shows the ESP32 Development Board, which is the general-purpose version of the ESP32 without a built-in camera module like the ESP32-CAM. It is widely used in IoT and embedded projects due to its processing capability, low cost, and ease of programming, making it one of the most popular microcontroller boards for developers. The ESP32 serves as the central communication and control unit that receives commands from the gesture recognition interface and wirelessly transmits the appropriate signals to the relay module, enabling the switching of connected appliances in real time.



Figure 6. *Relay Module (1 channel, 5v)*



Figure 7. *Contactor*

Figure 6 displays a relay module, which serves as an electrically controlled switch that allows a low-power device like an ESP32 to safely control high-voltage or high-current appliances. In operation, the ESP32 sends a small 3.3V logic signal to the relay's input pin, energizing an internal coil that switches the relay contacts, thereby connecting or disconnecting the external circuit. However, unlike industrial contactors, relay modules have current and voltage limitations, meaning they can only handle smaller household loads and are not suitable for controlling high-power appliances that require heavy-duty switching components. Figure 7 presents a contactor, a larger and more durable electrically operated switch designed to handle significantly higher power loads than a relay. Contactors are commonly used in industrial applications to control motors, HVAC systems, and other high-current equipment. In the context of the smart home system, a contactor would be used when controlling appliances that exceed the current rating of a standard relay module, ensuring safe operation and preventing electrical overheating or component damage. This makes the contactor an essential component for extending the system's capability to manage heavier household or commercial devices.



Figure 8. *Micro USB Cable*



Figure 9. *Power Adapter*

Figure 8 shows a Micro USB cable, a critical hardware component for the Smart Home Automation System with Hand Gesture Recognition. It serves two essential functions: providing power to the ESP32 Development Board and the ESP32-CAM

module and enabling stable operation of both microcontroller units by delivering the required voltage and current for processing, communication, and gesture recognition tasks. Figure 9 displays a power adapter, which converts AC power from a wall outlet into DC power suitable for electronic devices. In this system, the adapter works in conjunction with the Micro USB cable, supplying the necessary power to both the ESP32 and ESP32-CAM modules. This combination ensures that the microcontrollers receive a consistent and reliable power source, enabling uninterrupted operation of the gesture recognition and device control functions.



Figure 10. Extension Cord

Figure 10 shows an extension cord, commonly used for low-voltage applications such as powering radios, chargers, and small appliances. In the smart home system, each individual smart device uses an extension cord to consolidate power connections for all its components. For example, a single smart device may have an ESP32 microcontroller with its own power cable, a relay module with a separate power input, and in some cases, a contactor for high-voltage appliances that also requires a dedicated plug. The extension cord allows all these inputs to be connected conveniently to a single power source. Additionally, another extension cord is used as the output of the device, enabling connected appliances to receive power through the relay or contactor. This setup ensures organized and safe power

distribution for each smart device, simplifying connections while supporting both low- and high-voltage components within the system.

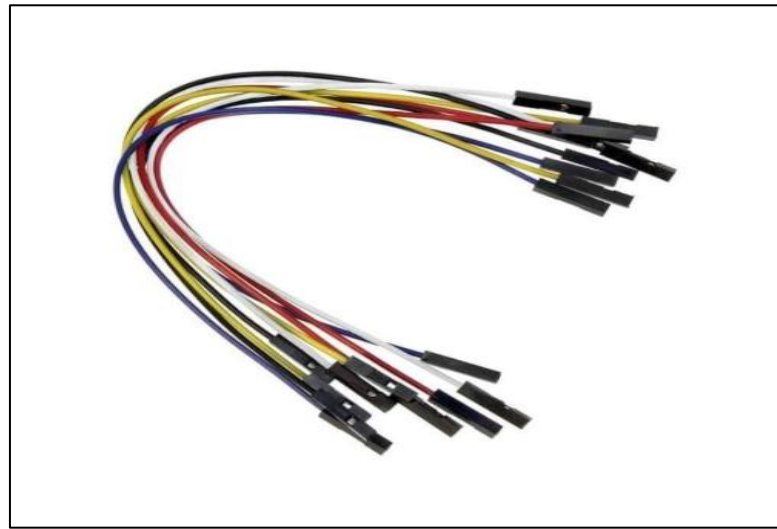


Figure 11. *Jumper Wires*

Figure 11 shows female-to-female jumper wires with color-coded connectors, commonly used to connect components or microcontroller pins. In the smart home system, they link the ESP32 microcontrollers to the relay modules, enabling control signals to switch appliances on or off while allowing easy, solder-free connections.

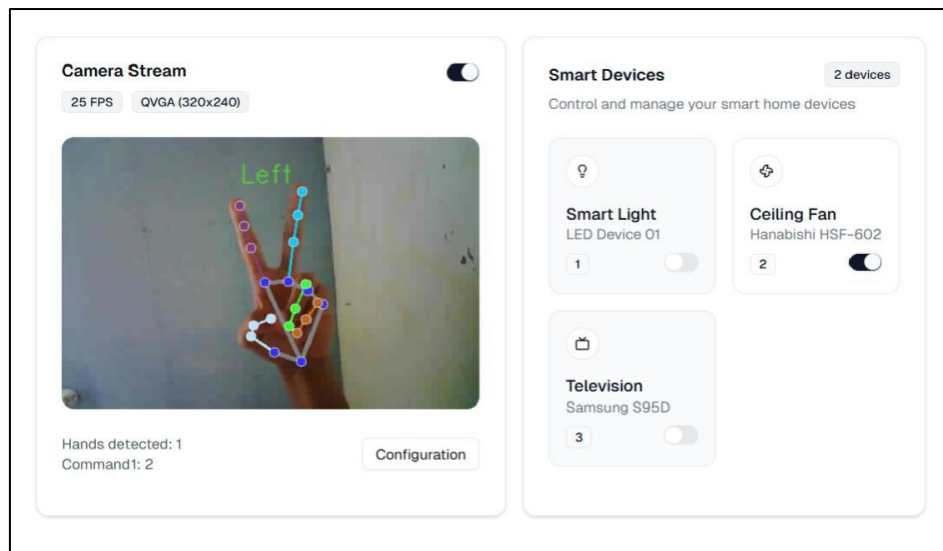


Figure 12. *User Interface*

This figure illustrates the main user interface (UI) and control dashboard of the Smart Home Automation System with Hand Gesture Recognition. The dashboard serves as the central hub where users can monitor and manage connected devices.

In addition to enabling intuitive gesture-based control, the UI provides real-time feedback, device status monitoring, and manual control options, ensuring both accessibility and usability. The system comprises three primary components. The first is the live camera feed displayed on the upper-left section of the interface, which serves as the visual input for the hand gesture recognition system. The process begins with the onboard camera capturing the raw video stream, which is then transmitted to the server for processing. Gesture recognition algorithms are applied to the stream, and the processed frames are re-encoded for transmission. These frames are made available to the web application through HTTP streaming, allowing users to view the real-time feed. System messages, such as “Hands detected: 1,” provide immediate feedback on recognition results generated by the server.

The second component is the Smart Device Management Panel, located in the lower section of the interface. This panel functions as the central control hub for all connected smart devices. Each device is presented with a user-defined label and corresponding model information, ensuring clarity and ease of identification. Control options are available alongside each device listing, enabling users to toggle power states, adjust settings, or perform specific actions as needed.

The third component is the Smart Device Hardware, which represents the physical IoT unit responsible for bridging the software interface with real-world appliances. It is composed of an ESP32 microcontroller, a relay module, and an extension cord. Household appliances can be connected directly to the extension cord, while the relay—controlled by the ESP32—manages their power states. The ESP32 receives commands wirelessly, either from the gesture recognition system or through manual interactions with the UI, allowing seamless integration between digital control and physical devices.

System Evaluation Results and Interpretation

Table 4. *Summary Evaluation of Benchmark Test*

Description	Mean	Interpretation
FUNCTIONALITY		
The system accurately recognizes and processes hand gestures.	5.0	Excellent
The system successfully executes commands without errors.	5.0	Excellent
The system provides appropriate feedback when a command is processed.	5.0	Excellent
RELIABILITY		
The system performs consistently without unexpected failures.	5.0	Excellent
The system maintains performance even with multiple users in the frame.	5.0	Excellent
The system remains stable even after prolonged usage.	5.0	Excellent
USABILITY		
The system is easy to learn and use for first-time users.	5.0	Excellent
The interface is intuitive and requires minimal effort to operate.	5.0	Excellent
The response time between gestures and system execution is reasonable.	5.0	Excellent
MAINTAINABILITY		
The system can be easily fixed or restored when problems occur.	5.0	Excellent
The system continues to work properly after being restarted.	5.0	Excellent
The system structure is organized in a way that makes troubleshooting simple.	5.0	Excellent
Overall Mean	5.0	Excellent

Table 4 shows the overall results of the benchmark test from the items in the survey questionnaire, including the overall weighted mean of 5 and verbal interpretation of "Excellent."

Table 5. *Distribution of the responses of the researchers on Benchmark Test.*

Category	Frequency	Percentage
Excellent	3	100%
Very Good	0	0%
Good	0	0%
Fair	0	0%
Poor	0	0%
Total	3	100%

Table 5 shows that all indicators across functionality, reliability, usability, and maintainability were rated Excellent by the researchers, with no responses in the lower categories. This indicates that the system fully satisfied the benchmark standards.

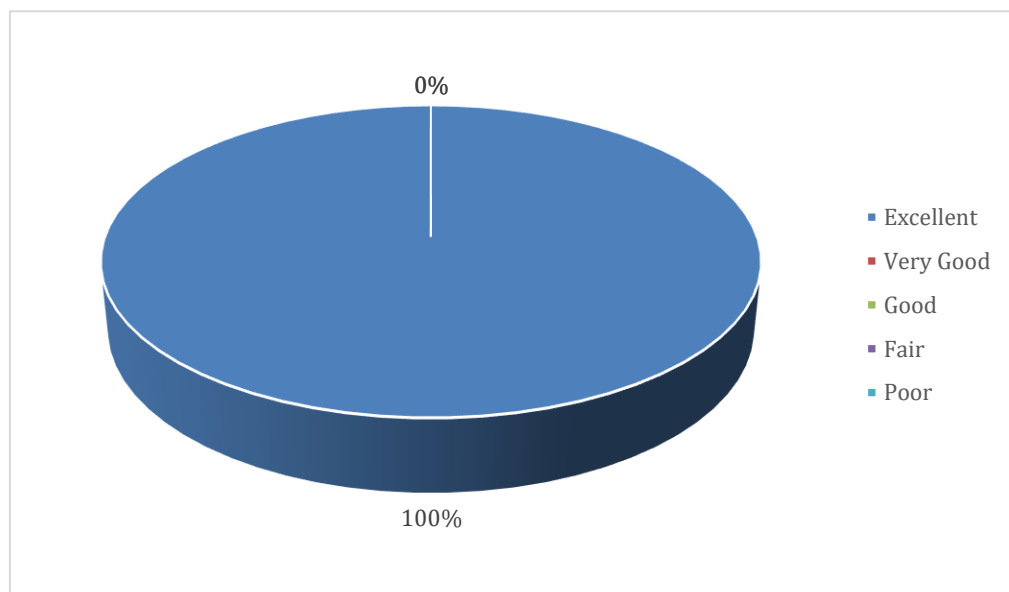
Figure 12. *Pie Chart for Benchmark Test*

Figure 12 presents the pie chart of the benchmark test results. The chart indicates that 100% of the responses fall under the “Excellent” category, with no ratings recorded in the lower categories. This outcome highlights the consistency of researcher evaluations and demonstrates that the system fully satisfied the established benchmark criteria.

Table 6. *Summary Evaluation of Pilot Test*

Description	Mean	Interpretation
FUNCTIONALITY		
The system accurately recognizes and processes hand gestures.	4.93	Excellent
The system successfully executes commands without errors.	4.83	Excellent
The system provides appropriate feedback when a command is processed.	4.83	Excellent
RELIABILITY		
The system performs consistently without unexpected failures.	4.80	Excellent
The system maintains performance even with multiple users in the frame.	4.73	Excellent
The system remains stable even after prolonged usage.	4.80	Excellent
USABILITY		
The system is easy to learn and use for first-time users.	4.80	Excellent
The interface is intuitive and requires minimal effort to operate.	4.83	Excellent
The response time between gestures and system execution is reasonable.	4.93	Excellent
MAINTAINABILITY		
The system can be easily fixed or restored when problems occur.	4.83	Excellent
The system continues to work properly after being restarted.	4.87	Excellent
The system structure is organized in a way that makes troubleshooting simple.	4.80	Excellent
Overall Mean	4.83	Excellent

Table 6 shows the overall results of the pilot test from the items in the survey questionnaire, including the overall weighted mean of 4.83 and verbal interpretation of "Excellent".

Table 7. *Distribution of the responses of the researchers on Pilot Test.*

Category	Frequency	Percentage
Excellent	25.67	85.56%
Very Good	3.83	12.78%
Good	0.50	1.67%
Fair	0	0%
Poor	0	0%
Total	30	100%

Table 7 shows that all indicators across functionality, reliability, usability, and maintainability were rated *Excellent* by the researchers, with no responses recorded in the lower categories. This result clearly indicates that the system fully satisfied the established benchmark standards.

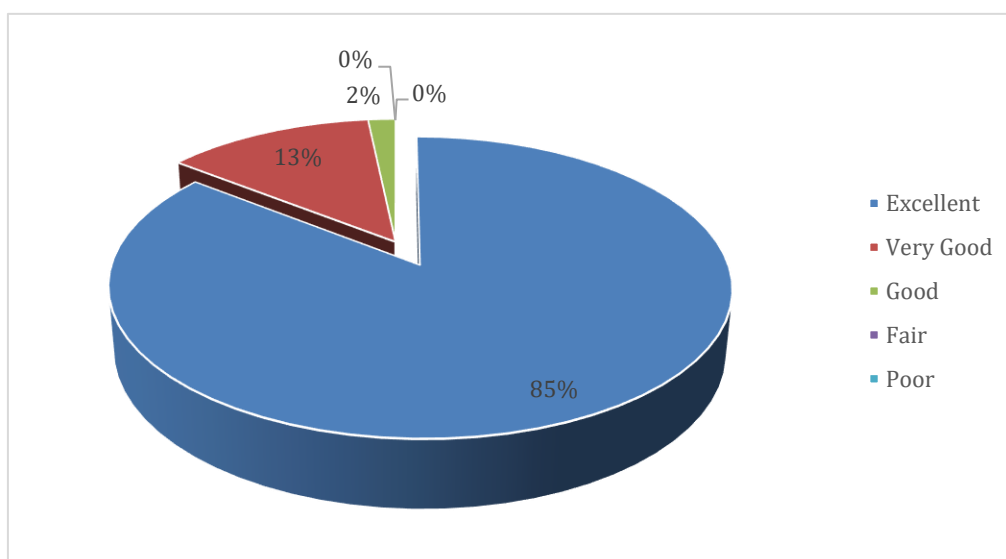


Figure 13. Pie Chart for Pilot Test

Figure 13 presents the pilot test results in a pie chart. The majority of responses rated the system as Excellent (85.56%), followed by Very Good (12.78%) and Good (1.67%), with no ratings under *Fair* or *Poor*. These findings indicate that IT experts found the system to be highly effective, reliable, and satisfactory.

Table 8. *Summary Evaluation of Final Test*

Description	Mean	Interpretation
FUNCTIONALITY		
The system accurately recognizes and processes hand gestures.	4.90	Excellent
The system successfully executes commands without errors.	4.86	Excellent
The system provides appropriate feedback when a command is processed.	4.86	Excellent
RELIABILITY		
The system performs consistently without unexpected failures.	4.90	Excellent
The system maintains performance even with multiple users in the frame.	4.83	Excellent
The system remains stable even after prolonged usage.	5.00	Excellent
USABILITY		
The system is easy to learn and use for first-time users.	4.93	Excellent
The interface is intuitive and requires minimal effort to operate.	4.80	Excellent
The response time between gestures and system execution is reasonable.	4.86	Excellent
MAINTAINABILITY		
The system can be easily fixed or restored when problems occur.	4.90	Excellent
The system continues to work properly after being restarted.	4.86	Excellent
The system structure is organized in a way that makes troubleshooting simple.	4.86	Excellent
Overall Mean	4.88	Excellent

Table 8 shows the overall results of the pilot test from the items in the survey questionnaire, including the overall weighted mean of 4.88 and verbal interpretation of "Excellent".

Table 9. *Distribution of the responses of the researchers on Final Test.*

Category	Frequency	Percentage
Excellent	25.8	86%
Very Good	4.30	14%
Good	0	0%
Fair	0	0%
Poor	0	0%
Total	30	100%

Table 9 presents the distribution of responses from the Final Test. The results indicate that most respondents rated the system as Excellent (86%), while the remaining 14% rated it as Very Good. No responses fell under Good, Fair, or Poor, indicating that the system consistently met users' expectations with a high level of performance and reliability.

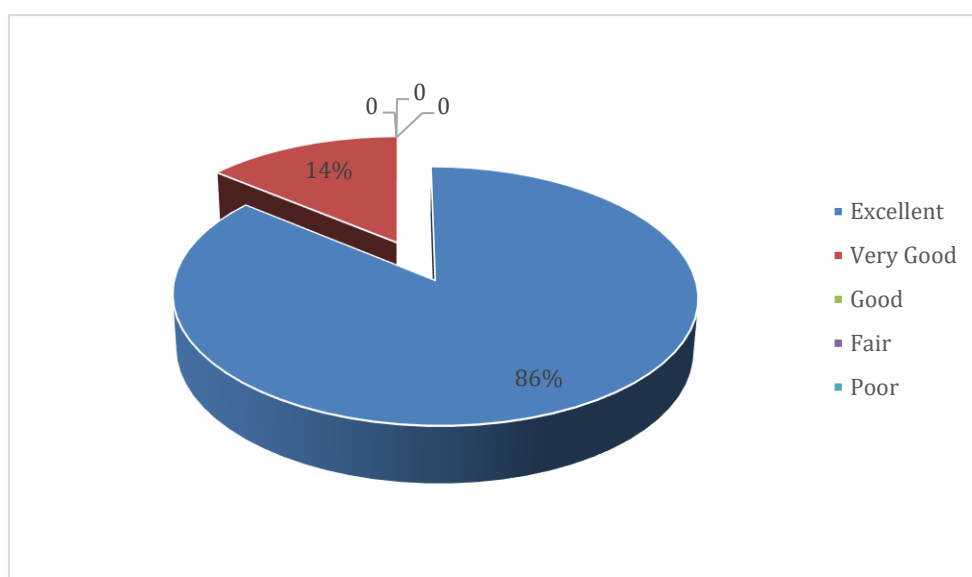
Figure 14. *Pie Chart for Final Test*

Figure 14 presents the final test results in a pie chart. The majority of responses rated the system as Excellent (86%) and followed by Very Good (14%), with no ratings under lower categories. These findings indicate that IT experts found the system to be highly effective, reliable, and satisfactory.

Table 10. *Percentage Distribution of the Responses of all Respondents*

Category	Mean	Interpretation
Benchmark Test	5.00	Highly Acceptable
Pilot Test	4.83	Highly Acceptable
Final Test	4.88	Highly Acceptable
Overall Mean	4.90	Highly Acceptable

The overall mean results across the three results indicate a score of 4.90, indicating a classification of “Highly Acceptable.”

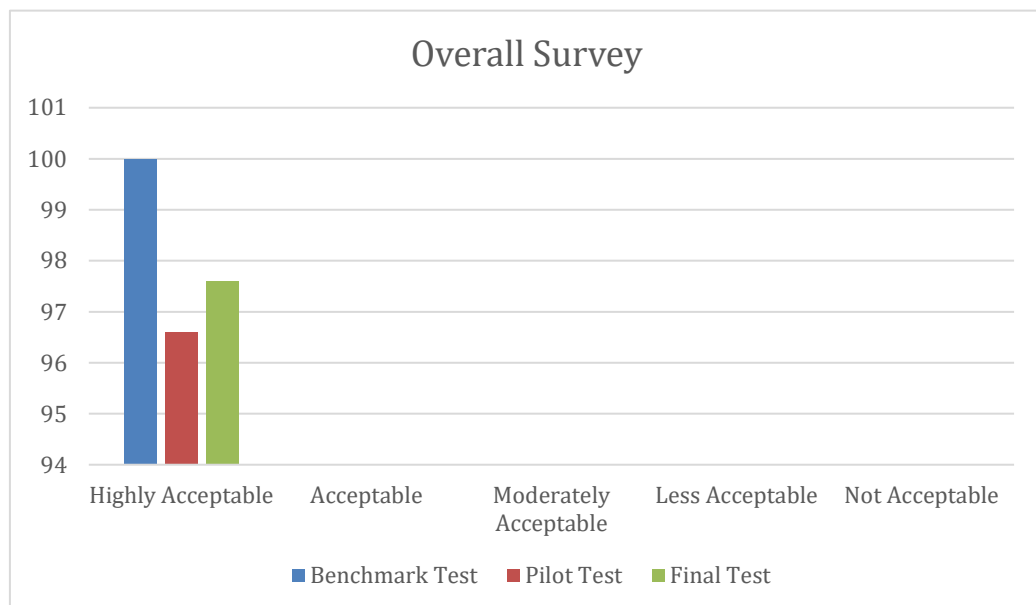
Figure 15. *Overall Survey Questionnaire Interpretation*

Figure 15 presented the overall survey results from the study's respondents. The graph provided a complete comparison of all survey data to assess whether the Smart Home Automation System, was rated as Highly Acceptable by respondents. It clearly showed that across all testing phases—benchmark, pilot, and final—the system consistently achieved high acceptance levels, reflecting its reliability and effectiveness as a learning tool. These results further confirmed the success of the system's improvements throughout the development process.

Users' SUS score

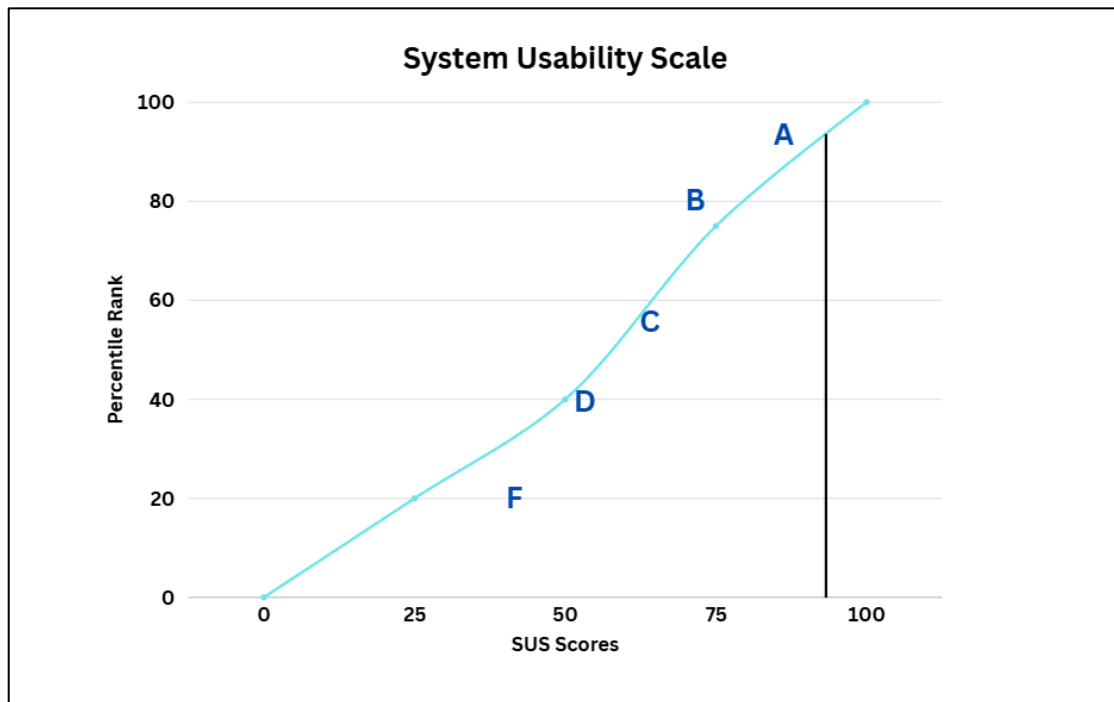


Figure 16. *Users' SUS score*

The graph above (Figure 18) illustrates the System Usability Scale (SUS) scores plotted against their corresponding percentile ranks, offering an overview of how users perceive the system's usability. The x-axis represents SUS scores (0–100), while the y-axis displays percentile ranks, showing how each score compares relative to others. SUS scores are grouped into letter grades: F (below 35) indicates poor usability, D (35–50) reflects below-average usability, C (50–70) suggests moderate or acceptable usability, B (70–85) represents good usability, and A (above 85) signifies excellent usability. With a SUS score of **92.83**, the system is categorized under the Excellent range (85–100) and corresponds to a grade of A. This result indicates that users perceived the system as highly usable, intuitive, and effective, reflecting a very positive user experience that supports its acceptance and potential adoption.

Chapter V

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter contains a summary of the study, findings, conclusion, and recommendations.

Summary of Findings

1. The design and development of the Smart Home Automation System with Hand Gesture Recognition successfully addressed the problem of limited accessibility, providing users, including those with mobility or speech impairments—a contactless way to control home appliances.
2. The system was evaluated in terms of functionality, reliability, usability, and maintainability through benchmark, pilot, and final testing phases. The benchmark test attained a perfect mean score of 5.0, indicating excellent performance. The pilot test achieved a mean score of 4.83, denoting satisfactory functionality, while the final test yielded a mean of 4.88, rated as Highly Acceptable. The overall mean of 4.90 confirmed that the system effectively met its objectives and performance standards.
3. The system achieved a usability score of 92.83 with a grade of “A” signifying highly acceptable performance.

Conclusion

Based on the findings, the conclusions below are presented:

1. The study concluded that the Smart Home Automation System with Hand Gesture Recognition effectively achieved its objectives by enabling users to control home devices efficiently and reliably. Its integration of the ESP32 module, relay, and extension cord allowed safe and responsive operation of connected appliances.
2. The system demonstrated strong performance in terms of functionality, reliability, usability, and maintainability, as reflected in the benchmark, pilot,

and final testing results. The benchmark test attained a mean score of 5.0 (Excellent), the pilot test achieved a mean of 4.83 (Excellent), and the final test yielded a mean of 4.88 (Excellent), resulting in an overall mean of 4.9 (Excellent). This indicates that the system consistently met the expected standards of quality and performance.

3. With the usability scale of 92.00 (Grade A), the system proved a highly acceptable level of usability, indicating that it is well-aligned with the needs of users by ensuring functionality, efficiency, and a satisfactory learning experience.

Recommendations

From the conclusions of the study, the researchers strongly recommend the following:

1. Future developers were encouraged to enhance the system by expanding the range of recognizable gestures and integrating additional smart devices for broader application.
2. Training sessions are recommended to ensure users can operate the system effectively. Brief orientation will help them become familiar with the hand gesture controls, minimize errors, and maximize ease of use and overall system efficiency, particularly for users with mobility or speech impairments.
3. Since the system attained a usability scale of 92.00 (Grade A), it is recommended that further refinements be made to improve its usability even further. Continuous testing and feedback from users, particularly those with mobility or speech impairments, should be conducted to ensure the system remains accessible, intuitive, and responsive to their needs.

Chapter VI

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APPENDICES



**EASTERN SAMAR
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August 27, 2025

JARED HAREM Q. CELIS

Campus Administrator
Eastern Samar State University Guiuan Campus
Brgy. Salug Guiuan, Eastern Samar

Dear Sir,

We, the undersigned fourth-year Bachelor of Science in Information Technology students, are currently conducting research titled “**Smart Home Automation System with Hand Gesture Recognition**” In line with this, we respectfully seek permission to distribute a survey questionnaire to selected faculty members on this campus as part of our study.

This research is a requirement for the partial fulfillment of our **Bachelor of Science in Information Technology** program, specifically under our **Capstone Project**.

The primary objective of this study is to collect relevant data regarding our research titled “**Smart Home Automation System with Hand Gesture Recognition**”. Please be assured that all gathered information will be kept strictly confidential and used solely for academic purposes.

We sincerely appreciate your time and consideration and look forward to your kind approval. Thank you for your support.

Respectfully,

JERSON D. CAIBOG

JEAN DYL M. ARENAJO

JESTHER O. MORALES

Recommended/Noted by:

ENGR. ROGER R. ABULENCIA

Dean, College of Computer Studies/Capstone Project Adviser

Action Taken:

☐ Approved

☐ Disapproved

JARED HAREM Q. CELIS, MIT
Campus Administrator



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August 27, 2025

ENGR. ROGER R. ABULENCIA

Dean, College of Computer Studies
Eastern Samar State University Guiuan Campus
Brgy. Salug Guiuan, Eastern Samar

Dear Sir:

Greetings!

We, the 4th-year undergraduate students taking up a Bachelor of Science in Information Technology of **Eastern Samar State University - Guiuan Campus** respectfully ask permission from your office to administer a survey questionnaire with the chosen IT experts and respondents regarding on the **Smart Home Automation System with Hand Gesture Recognition**.

This study is one of the requirements for the partial fulfillment of the program **Bachelor of Science in Information Technology**, particularly in our **Capstone Project**.

Rest assured that all data gathered will be treated with strict confidentiality.

We are anticipating your kind approval on this request. Thank you very much for your kind assistance and support.

Respectfully,

JERSON D. CAIBOG

JEAN DYL M. ARENAJO

JESTHER O. MORALES

Researchers

Action Taken:

☐ Approved

☐ Disapproved

ENGR. ROGER R. ABULENCIA

Dean, College of Computer Studies/Capstone Adviser



August 27, 2025

Dear Respondent:

The undersigned is presently undertaking a research study entitled: **"SMART HOME AUTOMATION SYSTEM WITH HAND GESTURE RECOGNITION"**

Along this, you were chosen as one of the respondents since you could absolutely give the needed information relevant to the present study.

Generated data or information from the questionnaire will enable the researcher to answer the specific problem set in this investigation.

Please feel free to answer the herein attached questionnaire as honestly as you can.

Rest assured that your responses would strictly be held in utmost confidentiality.

Thank you and more power.

Very truly yours,

JERSON CAIBOG

Research Leader

SURVEY QUESTIONNAIRE

Smart Home Automation System with Hand Gesture Recognition

Adopted from ISO/EIC 25010

Name: (optional) _____ Age: _____ date: _____

Please check if: ☐ IT Expert ☐ Homeowner

Direction: Evaluate the proposed system based on the following criteria: functionality, reliability, usability, and maintainability. Rate each item using the scale provided.

Legend: 5-Excellent 4-Very good 3-Good 2-Fair 1-Poor

CRITERIA	5	4	3	2	1
1. Functionality					
1.1 The system accurately recognizes and processes hand gestures.					
1.2 The system successfully executes commands without errors.					
1.3 The system provides appropriate feedback when a command is processed.					
2. Reliability					
2.1 The system performs consistently without unexpected failures.					
2.2 The system maintains performance even with multiple users in the frame.					
2.3 The system remains stable even after prolonged usage.					
3. Usability					
3.1 The system is easy to learn and use for first-time users.					
3.2 The interface is intuitive and requires minimal effort to operate.					
3.3 The response time between gestures and system execution is reasonable.					
4. Maintainability					
3.1 The system can be easily fixed or restored when problems occur.					
3.2 The system continues to work properly after being restarted.					
3.3 The system structure is organized in a way that makes troubleshooting simple.					

Respondent's signature

SYSTEM USABILITY SCALE

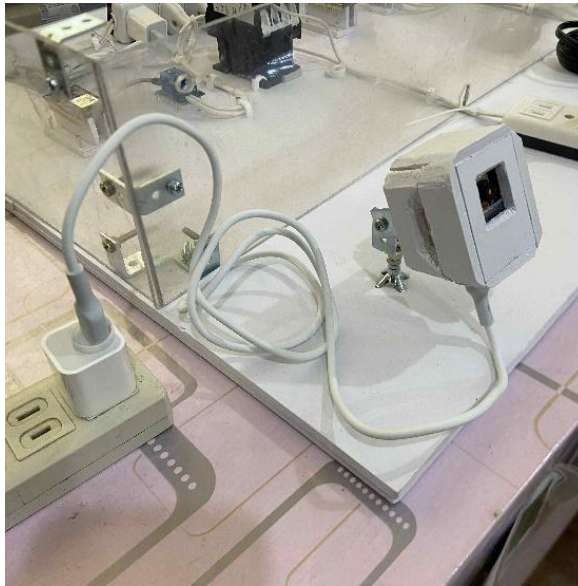
Student ID: _____

Date: _____

Instruction: For each of the following statements, mark one box that best describes your reactions to the Smart Home Automation System with Hand Gesture Recognition.

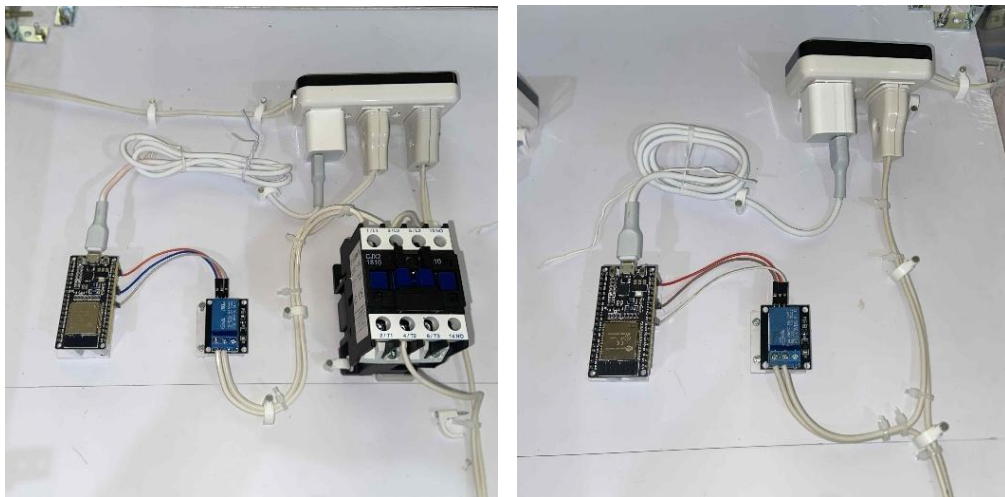
		Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
1	I think that I would like to use this system frequently					
2	I found this system unnecessarily complex					
3	I thought this system was easy to use					
4	I think that I would need assistance to be able to use system					
5	I found the various functions in this system were well-integrated					
6	I thought there was too much inconsistency in this system					
7	I would imagine that most people would learn to use this system very quickly					
8	I found this system very cumbersome (hard) to use					
9	I felt confident using this system					
10	I need to learn a lot of things before I can get going with system					

USER MANUAL



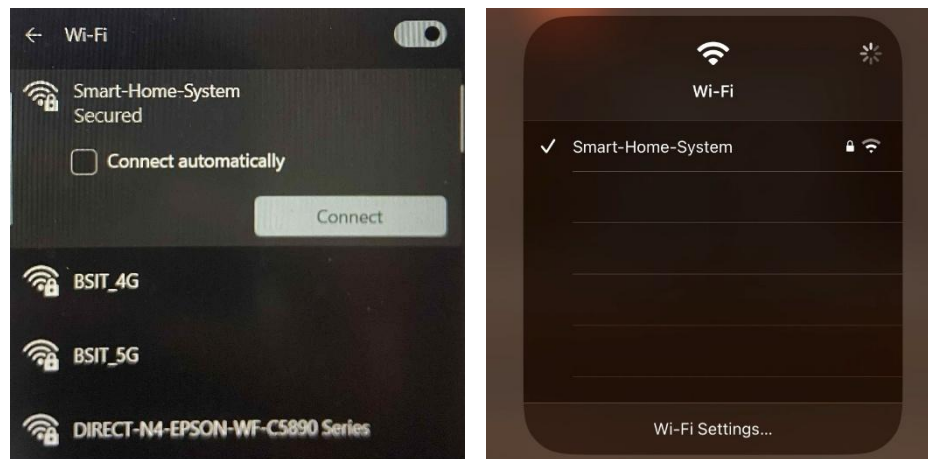
Plug in the Camera Module

Connect the camera module to your main controller, such as your Raspberry Pi or desktop computer. Make sure it is securely attached and properly recognized by the system. The camera is used to detect hand gestures for controlling your devices.



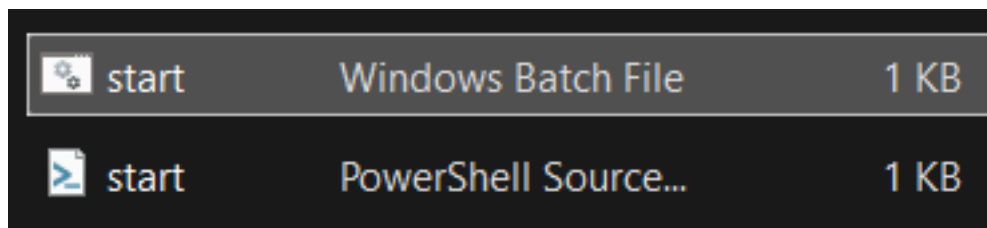
Plug in All Smart Devices

Connect all your smart devices, such as lights, fans, or sensors, to a power source. Make sure they are within range of the system network so they can connect automatically once the system is running.



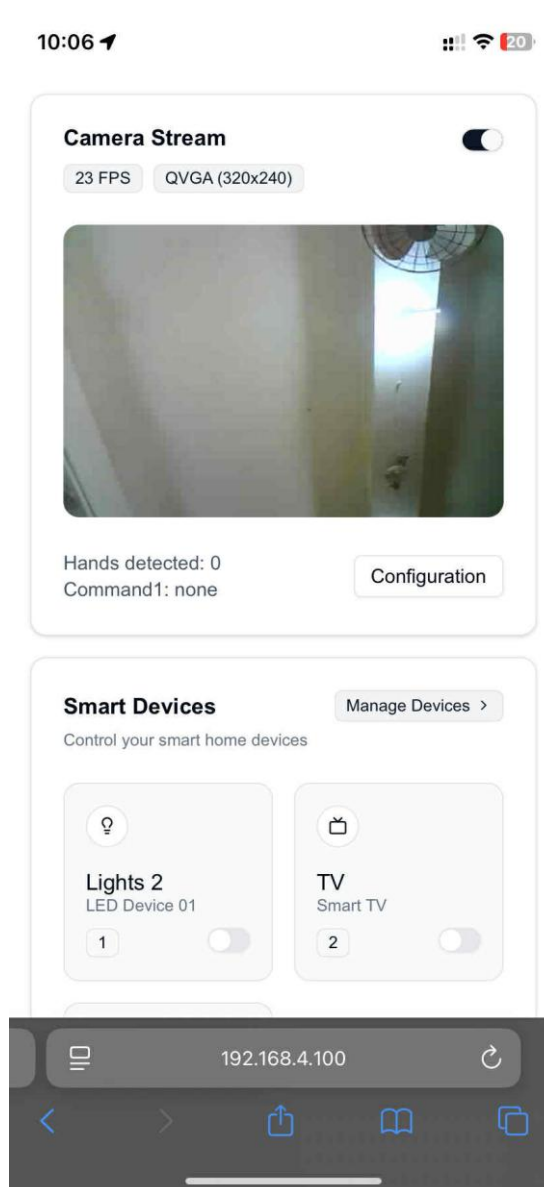
Connect to the System Wi-Fi

On your computer and mobile device (optional) go to your Wi-Fi settings and connect to the network named “Smart-Home-System.” You do not need an internet connection for this; it is only used for communication between your device and the system.



Run the Software

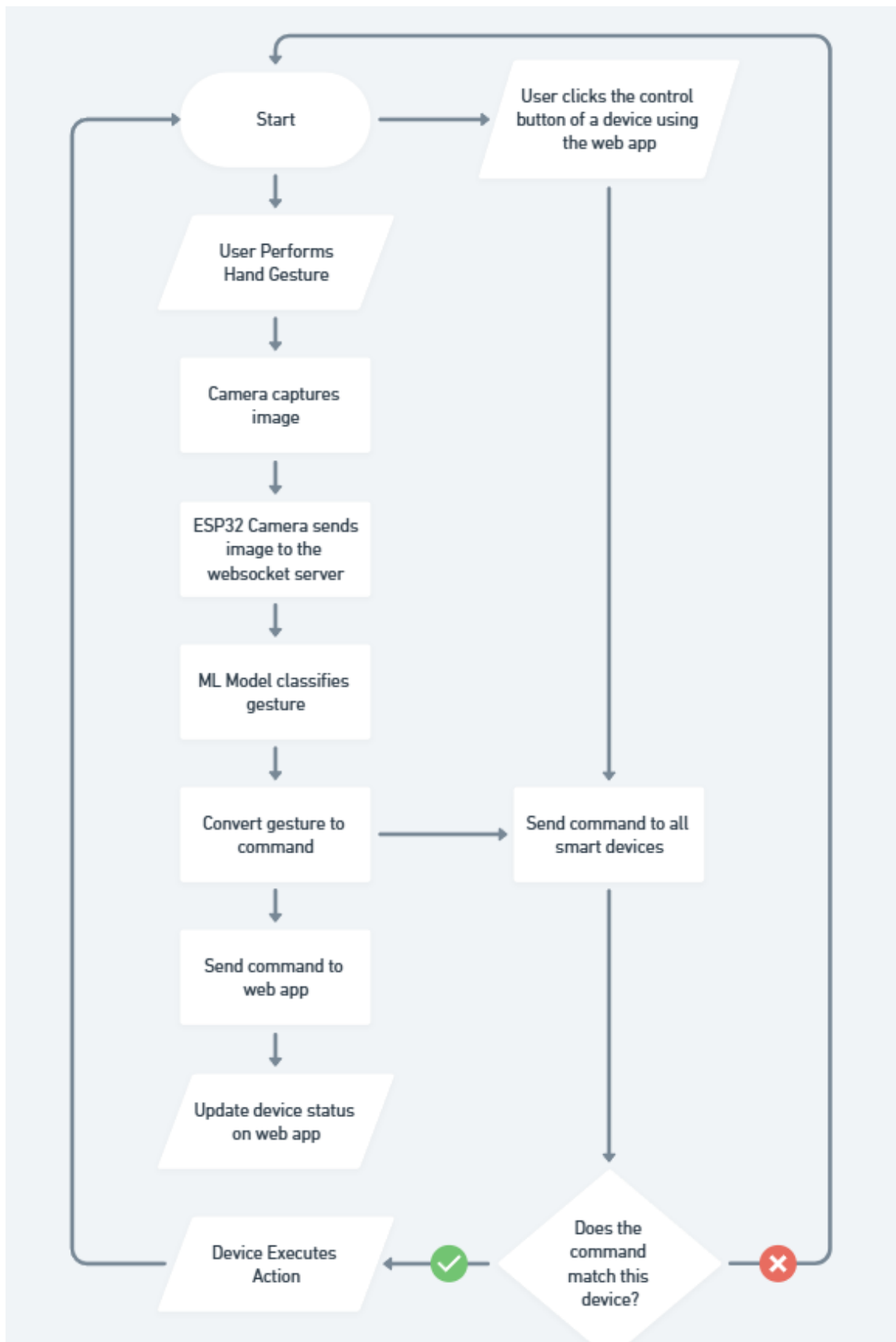
On your computer, find and double-click the batch (.bat) file named “start.bat”. This will run a PowerShell Source File (.ps1) named “start.ps1”, which starts all the necessary services for the system. Wait a few seconds for it to fully initialize. Once running, the system will automatically recognize hand gestures and control the connected devices.



Open the Web App (Optional)

To monitor or manually control your devices, open a web browser and type 192.168.4.100:3000 in the address bar. This will open the Smart Home System web app. You can use it to view device activity and control them manually. This step is optional; the system will still function even if you don't open the web app.

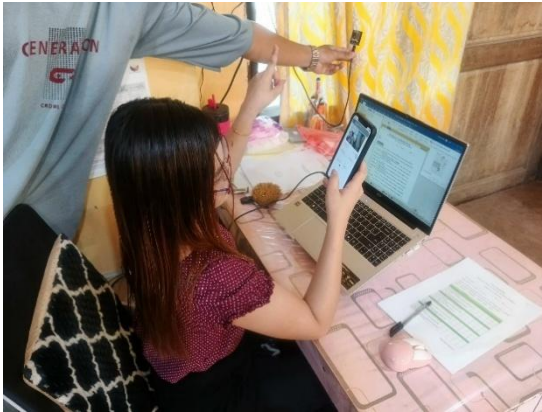
SYSTEM FLOWCHART



GANTT CHART

[illegible]

DOCUMENTATION





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EDITOR`S CERTIFICATION

This is to certify that this research study entitled **“SMART HOME AUTOMATION SYSTEM WITH HAND GESTURE RECOGNITION”**, prepared and submitted by **JERSON D. CAIBOG, JEAN DYL M. ARENAJO, AND JESTHER O. MORALES.**, in partial fulfillment of the requirements for the **DEGREE BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**, has been edited and reviewed by the undersigned.

ANGELO GABRIEL I. LAGRAMADA
Editor



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PROOFREADER'S CERTIFICATION

This is to certify that the thesis entitled, “**SMART HOME AUTOMATION SYSTEM WITH HAND GESTURE RECOGNITION**” prepared by Jerson D. Caibog et.al in partial fulfillment of the requirements for the **Degree Bachelor of Science in Information Technology** has been proofread by the undersigned.

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